Sources: Sheldon Axler, *Measure, Integration & Real Analysis* (MIRA) and its *Supplement*, at measure.axler.net.

Exercise 2.1 (MIRA 2C.1). Explain why there does not exist a measure space $(X, \mathcal{S}, \mu)$ with the property that $\{\mu(E) : E \in \mathcal{S}\} = [0, 1)$.

Solution 2.1. Suppose $(X, \mathcal{S}, \mu)$ is a measure space. Then $X \in \mathcal{S}$ because every σ-algebra on X contains X (2.25(a)), and every $E \in \mathcal{S}$ satisfies $E \subseteq X$ and hence $\mu(E) \le \mu(X)$, because measures preserve order (2.57(a)). Thus $\mu(X)$ is the largest element of $\{\mu(E) : E \in \mathcal{S}\}$. But $[0, 1)$ has no largest element: if $t \in [0, 1)$, then $\frac{1}{2}(t + 1) \in [0, 1)$ and $\frac{1}{2}(t + 1) > t$. Hence $\{\mu(E) : E \in \mathcal{S}\} \neq [0, 1)$. □

Exercise 2.2 (MIRA 2C.5). Suppose $(X, \mathcal{S}, \mu)$ is a measure space such that $\mu(X) < \infty$. Prove that if $\mathcal{A}$ is a set of disjoint sets in $\mathcal{S}$ such that $\mu(A) > 0$ for every $A \in \mathcal{A}$, then $\mathcal{A}$ is a countable set.

Solution 2.2. For $n \in \mathbf{Z}^+$, let $\mathcal{A}_n = \{A \in \mathcal{A} : \mu(A) > \frac{1}{n}\}$. Suppose $m \in \mathbf{Z}^+$ and $A_1, \dots, A_m$ are distinct elements of $\mathcal{A}_n$. They are disjoint sets in $\mathcal{S}$. Thus finite additivity of μ, which follows from the definition of measure (2.54), and order preservation (2.57(a)) give

$$\frac{m}{n} < \mu(A_1) + \cdots + \mu(A_m) = \mu(A_1 \cup \cdots \cup A_m) \le \mu(X).$$

Hence $m < n\mu(X) < \infty$. Thus $\mathcal{A}_n$ has at most $n\mu(X)$ elements and is finite.

Now suppose $A \in \mathcal{A}$. Then $0 < \mu(A) \le \mu(X) < \infty$ by 2.57(a), and the Archimedean Property gives $n \in \mathbf{Z}^+$ with $\frac{1}{n} < \mu(A)$. Thus $A \in \mathcal{A}_n$. Hence $\mathcal{A} = \bigcup_{n=1}^{\infty} \mathcal{A}_n$.

If $\mathcal{A} = \emptyset$, then $\mathcal{A}$ is countable. Thus assume $\mathcal{A} \neq \emptyset$. Because $\mathcal{A}_1 \subseteq \mathcal{A}_2 \subseteq \cdots$ and each $\mathcal{A}_n$ is finite, listing the elements of $\mathcal{A}_1$, then those of $\mathcal{A}_2 \setminus \mathcal{A}_1$, and so on, produces a sequence containing every element of $\mathcal{A}$. Thus $\mathcal{A}$ is countable. □

Exercise 2.3 (MIRA 2D.5). Prove that if $A \subseteq \mathbf{R}$ is Lebesgue measurable, then there exists an increasing sequence $F_1 \subseteq F_2 \subseteq \cdots$ of closed sets contained in A such that

$$\left| A \setminus \bigcup_{k=1}^{\infty} F_k \right| = 0.$$

Solution 2.3. Because A is Lebesgue measurable, the equivalences for being a Lebesgue measurable set (2.71) give closed sets $F_1, F_2, \dots$ contained in A such that $\left|A \setminus \bigcup_{k=1}^{\infty} F_k\right| = 0$. For $k \in \mathbf{Z}^+$, let $F'_k = F_1 \cup \cdots \cup F_k$. Then $F'_1 \subseteq F'_2 \subseteq \cdots$, and each F'_k is a subset of A that is closed because a finite union of closed sets is closed. Because $F_k \subseteq F'_k \subseteq \bigcup_{j=1}^{\infty} F_j$ for every k, we have $\bigcup_{k=1}^{\infty} F'_k = \bigcup_{k=1}^{\infty} F_k$. Hence

$$\left| A \setminus \bigcup_{k=1}^{\infty} F'_k \right| = \left| A \setminus \bigcup_{k=1}^{\infty} F_k \right| = 0,$$

and $F'_1 \subseteq F'_2 \subseteq \cdots$ is the desired sequence. □

Exercise 2.4 (MIRA 2D.7). Prove that if $A \subseteq \mathbf{R}$ is Lebesgue measurable, then there exists a decreasing sequence $G_1 \supseteq G_2 \supseteq \cdots$ of open sets containing A such that

$$\left|\left(\bigcap_{k=1}^{\infty} G_k\right) \setminus A\right| = 0.$$

Solution 2.4. Suppose A is Lebesgue measurable. By the equivalences for being a Lebesgue measurable set (2.71), there exist open sets $G_1, G_2, \ldots$ containing A such that $\left|\left(\bigcap_{k=1}^{\infty} G_k\right) \setminus A\right| = 0$. For $k \in \mathbf{Z}^+$, let $G'_k = G_1 \cap \cdots \cap G_k$, which contains A and is open because a finite intersection of open sets is open. Then $G'_1 \supseteq G'_2 \supseteq \cdots$, and $\bigcap_{j=1}^{\infty} G_j \subseteq G'_k \subseteq G_k$ for every k, which implies that $\bigcap_{k=1}^{\infty} G'_k = \bigcap_{k=1}^{\infty} G_k$. Thus $G'_1 \supseteq G'_2 \supseteq \cdots$ is a decreasing sequence of open sets containing A with

$$\left|\left(\bigcap_{k=1}^{\infty} G'_k\right) \setminus A\right| = \left|\left(\bigcap_{k=1}^{\infty} G_k\right) \setminus A\right| = 0. \qquad \square$$

Exercise 2.5 (MIRA 2D.10). Prove that if A and B are disjoint subsets of $\mathbf{R}$ and B is Lebesgue measurable, then $|A \cup B| = |A| + |B|$.

Solution 2.5. Countable subadditivity of outer measure (2.8) gives $|A \cup B| \leq |A| + |B|$. To prove the inequality in the other direction, recall that B is Lebesgue measurable. Thus there exists a Borel set $B' \subseteq B$ such that $|B \setminus B'| = 0$, by the definition of Lebesgue measurable set (2.70). Then $|B'| \leq |B|$ because outer measure preserves order (2.5), and 2.8 applied to $B = B' \cup (B \setminus B')$ gives $|B| \leq |B'| + |B \setminus B'| = |B'|$. Thus $|B'| = |B|$.

The sets A and B' are disjoint because $B' \subseteq B$ and $A \cap B = \emptyset$, and B' is a Borel set. Thus additivity of outer measure if one of the sets is a Borel set (2.66) gives $|A \cup B'| = |A| + |B'|$. Because $A \cup B' \subseteq A \cup B$, 2.5 now gives

$$|A \cup B| \geq |A \cup B'| = |A| + |B'| = |A| + |B|. \qquad \square$$

Exercise 2.6 (MIRA 2E.4). Prove or give a counterexample: If $A \subseteq \mathbf{R}$ and $f_1, f_2, \ldots$ is a sequence of uniformly continuous functions from A to $\mathbf{R}$ that converges uniformly to a function $f \colon A \to \mathbf{R}$, then f is uniformly continuous on A.

Solution 2.6. The statement is true. Suppose $\varepsilon > 0$. By the definition of uniform convergence (2.82), there exists $n \in \mathbf{Z}^+$ such that $|f_n(x) - f(x)| < \frac{\varepsilon}{3}$ for all $x \in A$. Because f_n is uniformly continuous on A, there exists $\delta > 0$ such that $|f_n(x) - f_n(y)| < \frac{\varepsilon}{3}$ for all $x, y \in A$ with $|x - y| < \delta$. Now suppose $x, y \in A$ and $|x - y| < \delta$. The triangle inequality gives

$$|f(x) - f(y)| \leq |f(x) - f_n(x)| + |f_n(x) - f_n(y)| + |f_n(y) - f(y)| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.$$

Thus f is uniformly continuous on A. $\square$

Exercise 2.7 (MIRA 2E.5). Give an example to show that Egorov's Theorem can fail without the hypothesis that $\mu(X) < \infty$.

Solution 2.7. Let $X = \mathbf{Z}^+$, let $\mathcal{S}$ be the σ-algebra of all subsets of $\mathbf{Z}^+$, and let μ be counting measure (Example 2.55) on $(X, \mathcal{S})$. Then $\mu(X) = \infty$ because $\mathbf{Z}^+$ is not a finite set. For $k \in \mathbf{Z}^+$, let

$f_k = \chi_{\{k\}}$, and let $f\colon \mathbf{Z}^+ \to \mathbf{R}$ be the function 0. Every function from $\mathbf{Z}^+$ to $\mathbf{R}$ is $\mathcal{S}$-measurable because $\mathcal{S}$ contains every subset of $\mathbf{Z}^+$. If $x \in \mathbf{Z}^+$, then $f_k(x) = 0$ for all integers $k > x$, and thus $\lim_{k\to\infty} f_k(x) = 0 = f(x)$. Hence every hypothesis of Egorov's Theorem (2.85) holds except $\mu(X) < \infty$.

The conclusion of 2.85 fails for $\varepsilon = 1$. Suppose $E \in \mathcal{S}$ and $\mu(X \setminus E) < 1$. A nonempty subset of $\mathbf{Z}^+$ has counting measure at least 1, because it is either a finite set with at least one element or not a finite set. Thus $X \setminus E = \emptyset$, which implies that $E = \mathbf{Z}^+$. But $f_1, f_2, \ldots$ does not converge uniformly to f on $\mathbf{Z}^+$: for every $n \in \mathbf{Z}^+$ we have $|f_n(n) - f(n)| = 1$, and thus no $n \in \mathbf{Z}^+$ satisfies the definition of uniform convergence (2.82) with $\frac{1}{2}$ in place of ε. □

Exercise 2.8 (MIRA 2E.7). Suppose F is a closed bounded subset of $\mathbf{R}$ and $g_1, g_2, \ldots$ is an increasing sequence of continuous real-valued functions on F (thus $g_1(x) \le g_2(x) \le \cdots$ for all $x \in F$) such that $\sup\{g_1(x), g_2(x), \ldots\} < \infty$ for each $x \in F$. Define a real-valued function g on F by

$$g(x) = \lim_{k\to\infty} g_k(x).$$

Prove that g is continuous on F if and only if $g_1, g_2, \ldots$ converges uniformly on F to g.
[*The result above is called Dini's Theorem.*]

Solution 2.8. If $F = \emptyset$, then both conditions hold. Thus assume $F \neq \emptyset$. For each $x \in F$, the sequence $g_1(x), g_2(x), \ldots$ is increasing and bounded above by $\sup\{g_1(x), g_2(x), \ldots\} < \infty$, and thus it converges because bounded monotone sequences converge; hence $g(x) \in \mathbf{R}$. Also $g_k(x) \le g(x)$ for all $k \in \mathbf{Z}^+$: if $g_k(x) > g(x)$, then $g_j(x) \ge g_k(x) > g(x)$ for all $j \ge k$, and thus $|g_j(x) - g(x)| \ge g_k(x) - g(x) > 0$ for all $j \ge k$, which contradicts $\lim_{j\to\infty} g_j(x) = g(x)$.

First suppose $g_1, g_2, \ldots$ converges uniformly on F to g. Because each g_k is continuous, the uniform limit of continuous functions is continuous (2.84), and thus g is continuous on F.

To prove the other direction, suppose g is continuous on F and $\varepsilon > 0$. For each $x \in F$, because $\lim_{k\to\infty} g_k(x) = g(x)$, there exists $k_x \in \mathbf{Z}^+$ such that $g(x) - g_{k_x}(x) < \frac{\varepsilon}{2}$. Because g and g_{k_x} are continuous at x, there exists $\delta_x > 0$ such that $|g(y) - g(x)| < \frac{\varepsilon}{4}$ and $|g_{k_x}(y) - g_{k_x}(x)| < \frac{\varepsilon}{4}$ for all $y \in F$ with $|y - x| < \delta_x$. For such y we have

$$g(y) - g_{k_x}(y) \le |g(y) - g(x)| + \big(g(x) - g_{k_x}(x)\big) + |g_{k_x}(x) - g_{k_x}(y)| < \frac{\varepsilon}{4} + \frac{\varepsilon}{2} + \frac{\varepsilon}{4} = \varepsilon.$$

The open intervals $(x - \delta_x, x + \delta_x)$, for $x \in F$, form an open cover of F. Because F is a nonempty closed bounded set, the Heine–Borel Theorem (2.12) gives $m \in \mathbf{Z}^+$ and $x_1, \ldots, x_m \in F$ such that the intervals for $x_1, \ldots, x_m$ cover F. Let $N = \max\{k_{x_1}, \ldots, k_{x_m}\}$.

Now suppose $y \in F$ and $k \ge N$. Then $|y - x_j| < \delta_{x_j}$ for some $j \in \{1, \ldots, m\}$, and $k \ge k_{x_j}$. Because $g_1(y), g_2(y), \ldots$ is increasing, the estimate above gives $0 \le g(y) - g_k(y) \le g(y) - g_{k_{x_j}}(y) < \varepsilon$. Thus $|g_k(y) - g(y)| < \varepsilon$ for all integers $k \ge N$ and all $y \in F$. Hence $g_1, g_2, \ldots$ converges uniformly on F to g, completing the proof. □

Exercise 2.9 (MIRA 3A.7). Suppose X is a set, $\mathcal{S}$ is the σ-algebra of all subsets of X, and $w\colon X \to [0, \infty]$ is a function. Define a measure μ on $(X, \mathcal{S})$ by

$$\mu(E) = \sum_{x\in E} w(x)$$

for $E \subseteq X$. Prove that if $f\colon X \to [0,\infty]$ is a function, then

$$\int f\,d\mu = \sum_{x\in X} w(x)f(x),$$

where the infinite sums above are defined as the supremum of all sums over finite subsets of E (first sum) or X (second sum).

Solution 2.9. Because every subset of X is in $\mathcal{S}$, the function f is $\mathcal{S}$-measurable, and thus $\int f\,d\mu$ is defined. If $x \in X$, then the finite subsets of $\{x\}$ are $\emptyset$ and $\{x\}$, and thus $\mu(\{x\}) = w(x)$. Let $T = \sum_{x\in X} w(x)f(x)$, where $0\cdot\infty = 0$.

First we show that $T \le \int f\,d\mu$. Suppose D is a finite subset of X. If D is empty, then the sum $\sum_{x\in D} w(x)f(x)$ is 0, which is at most $\int f\,d\mu$. Otherwise write $D = \{x_1,\dots,x_n\}$ with $x_1,\dots,x_n$ distinct, and let P be the $\mathcal{S}$-partition of X consisting of the sets $\{x_1\}$, …, $\{x_n\}$, together with $X\setminus D$ if $D \ne X$. The term of $\mathcal{L}(f,P)$ that belongs to $X\setminus D$ lies in $[0,\infty]$, and $\mu(\{x_j\}) = w(x_j)$ and $\inf_{\{x_j\}} f = f(x_j)$ for each j. Thus the definitions of the lower Lebesgue sum (3.2) and of the integral (3.3) give

$$\int f\,d\mu \ge \mathcal{L}(f,P) \ge \sum_{j=1}^{n} w(x_j)f(x_j) = \sum_{x\in D} w(x)f(x).$$

Taking the supremum over all finite sets $D \subseteq X$ gives $T \le \int f\,d\mu$.

To prove the inequality in the other direction, suppose $A_1,\dots,A_m$ are disjoint subsets of X and $c_1,\dots,c_m \in [0,\infty)$ are such that $f \ge \sum_{j=1}^m c_j\chi_{A_j}$ on X. By the computation of integrals via finite simple functions (3.9), it suffices to show that $\sum_{j=1}^m c_j\mu(A_j) \le T$. If $T = \infty$, this holds; thus assume $T < \infty$. Deleting the terms with $c_j = 0$ changes neither $\sum_{j=1}^m c_j\chi_{A_j}$ nor $\sum_{j=1}^m c_j\mu(A_j)$, because $0\cdot\infty = 0$. Thus we can assume that $c_1,\dots,c_m \in (0,\infty)$.

If $x \in A_j$, then $x \notin A_i$ for $i \ne j$ because the sets are disjoint, and thus $f(x) \ge c_j$. Hence $c_j w(x) \le w(x)f(x)$ for all $x \in A_j$, where both sides equal ∞ if $w(x) = \infty$ because $c_j > 0$. Now suppose $D_1 \subseteq A_1,\dots,D_m \subseteq A_m$ are finite sets. They are disjoint, and their union D is a finite subset of X. Thus

$$\sum_{j=1}^{m} c_j \sum_{x\in D_j} w(x) = \sum_{j=1}^{m}\sum_{x\in D_j} c_j w(x) \le \sum_{x\in D} w(x)f(x) \le T.$$

Taking $D_i = \emptyset$ for $i \ne j$ shows that $\sum_{x\in D_j} w(x) \le T/c_j$ for every finite set $D_j \subseteq A_j$, and thus $\mu(A_j) \le T/c_j < \infty$ by the definition of μ. Suppose $\varepsilon > 0$. For each j, because $\mu(A_j)$ is the supremum of the sums $\sum_{x\in D_j} w(x)$ over finite sets $D_j \subseteq A_j$, the number $\mu(A_j) - \frac{\varepsilon}{mc_j}$ is not an upper bound of these sums; thus there is a finite set $D_j \subseteq A_j$ such that $c_j\sum_{x\in D_j} w(x) > c_j\mu(A_j) - \frac{\varepsilon}{m}$. Summing over j and using the inequality above gives $\sum_{j=1}^m c_j\mu(A_j) - \varepsilon < T$. Because ε is an arbitrary positive number, $\sum_{j=1}^m c_j\mu(A_j) \le T$. Now 3.9 implies that $\int f\,d\mu \le T$, completing the proof. □

Exercise 2.10 (MIRA 3A.17). *For $x_1, x_2, \dots$ a sequence in $[-\infty,\infty]$, define $\liminf_{k\to\infty} x_k$ by*

$$\liminf_{k\to\infty} x_k = \lim_{k\to\infty} \inf\{x_k, x_{k+1}, \dots\}.$$

Note that $\inf\{x_k, x_{k+1}, \dots\}$ is an increasing function of k; thus the limit above on the right exists in $[-\infty,\infty]$.

Suppose that $(X, \mathcal{S}, \mu)$ is a measure space and $f_1, f_2, \dots$ is a sequence of nonnegative $\mathcal{S}$-measurable functions on X. Define a function $f\colon X \to [0, \infty]$ by $f(x) = \liminf_{k\to\infty} f_k(x)$.

(a) Show that f is an $\mathcal{S}$-measurable function.

(b) Prove that

$$\int f \, d\mu \le \liminf_{k\to\infty} \int f_k \, d\mu.$$

(c) Give an example showing that the inequality in (b) can be a strict inequality even when $\mu(X) < \infty$ and the family of functions $\{f_k\}_{k\in\mathbf{Z}^+}$ is uniformly bounded.

[*The result in (b) is called Fatou's Lemma. Some textbooks prove Fatou's Lemma and then use it to prove the Monotone Convergence Theorem. Here we are taking the reverse approach—you should be able to use the Monotone Convergence Theorem to give a clean proof of Fatou's Lemma.*]

Solution 2.10. For $k \in \mathbf{Z}^+$, define $g_k\colon X \to [0, \infty]$ by $g_k(x) = \inf\{f_k(x), f_{k+1}(x), \dots\}$. If $x \in X$ and $k \in \mathbf{Z}^+$, then $g_k(x) \le g_{k+1}(x)$, because $\{f_{k+1}(x), f_{k+2}(x), \dots\}$ is a subset of $\{f_k(x), f_{k+1}(x), \dots\}$. Thus for each $x \in X$ we have $\lim_{k\to\infty} g_k(x) = \sup\{g_1(x), g_2(x), \dots\}$: if this supremum s equals ∞, then for every $t \in \mathbf{R}$ some $g_k(x)$ exceeds t and hence so does every later term; if $s < \infty$, then for every $\varepsilon > 0$ the number $s - \varepsilon$ is not an upper bound of $\{g_1(x), g_2(x), \dots\}$, and thus $s - \varepsilon < g_k(x) \le s$ for all sufficiently large k. Hence the definition of $\liminf$ gives $f(x) = \sup\{g_1(x), g_2(x), \dots\}$ for every $x \in X$.

(a) Each g_k is the infimum of the $\mathcal{S}$-measurable functions $f_k, f_{k+1}, \dots$, and thus g_k is $\mathcal{S}$-measurable by the result on the infimum and supremum of a sequence of $\mathcal{S}$-measurable functions (2.53). Hence $f = \sup\{g_1, g_2, \dots\}$ is $\mathcal{S}$-measurable by 2.53 again.

(b) Because $0 \le g_1 \le g_2 \le \cdots$ are $\mathcal{S}$-measurable and $\lim_{k\to\infty} g_k = f$ pointwise, the Monotone Convergence Theorem (3.11) gives $\int f \, d\mu = \lim_{k\to\infty} \int g_k \, d\mu$. Suppose $k \in \mathbf{Z}^+$. If $j \ge k$, then $g_k \le f_j$ on X, and thus $\int g_k \, d\mu \le \int f_j \, d\mu$ because integration is order preserving (3.8). Hence $\int g_k \, d\mu \le \inf\{\int f_k \, d\mu, \int f_{k+1} \, d\mu, \dots\}$, because $\int g_k \, d\mu$ is a lower bound of this set. The right side increases with k, and thus its limit exists and equals $\liminf_{k\to\infty} \int f_k \, d\mu$ by the definition of $\liminf$. Taking the limit as $k \to \infty$ of both sides of the inequality above gives

$$\int f \, d\mu = \lim_{k\to\infty} \int g_k \, d\mu \le \liminf_{k\to\infty} \int f_k \, d\mu,$$

as desired.

(c) Let μ be counting measure (Example 2.55) on $X = \{1, 2\}$ with $\mathcal{S}$ the σ-algebra of all subsets of X. Then $\mu(X) = 2 < \infty$. For $k \in \mathbf{Z}^+$, let $f_k = \chi_{\{1\}}$ if k is odd and $f_k = \chi_{\{2\}}$ if k is even. Each f_k is $\mathcal{S}$-measurable because every function from X to $\mathbf{R}$ is, and $0 \le f_k \le 1$ for every k; thus the family $\{f_k\}_{k\in\mathbf{Z}^+}$ is uniformly bounded.

Suppose $x \in X$. For every $k \in \mathbf{Z}^+$, one of k and $k + 1$ is odd and the other is even, and thus $f_k(x) = 0$ or $f_{k+1}(x) = 0$. Hence $\inf\{f_k(x), f_{k+1}(x), \dots\} = 0$ for every k, and the definition of $\liminf$ gives $f(x) = 0$. Thus $f = \chi_\emptyset$, and the integral of a characteristic function (3.4) gives $\int f \, d\mu = \mu(\emptyset) = 0$. Also, 3.4 gives $\int f_k \, d\mu = 1$ for every $k \in \mathbf{Z}^+$, because $\mu(\{1\}) = \mu(\{2\}) = 1$. Thus $\liminf_{k\to\infty} \int f_k \, d\mu = 1$, and the inequality in (b) is strict:

$$\int f \, d\mu = 0 < 1 = \liminf_{k\to\infty} \int f_k \, d\mu. \qquad \square$$